INCH-POUND

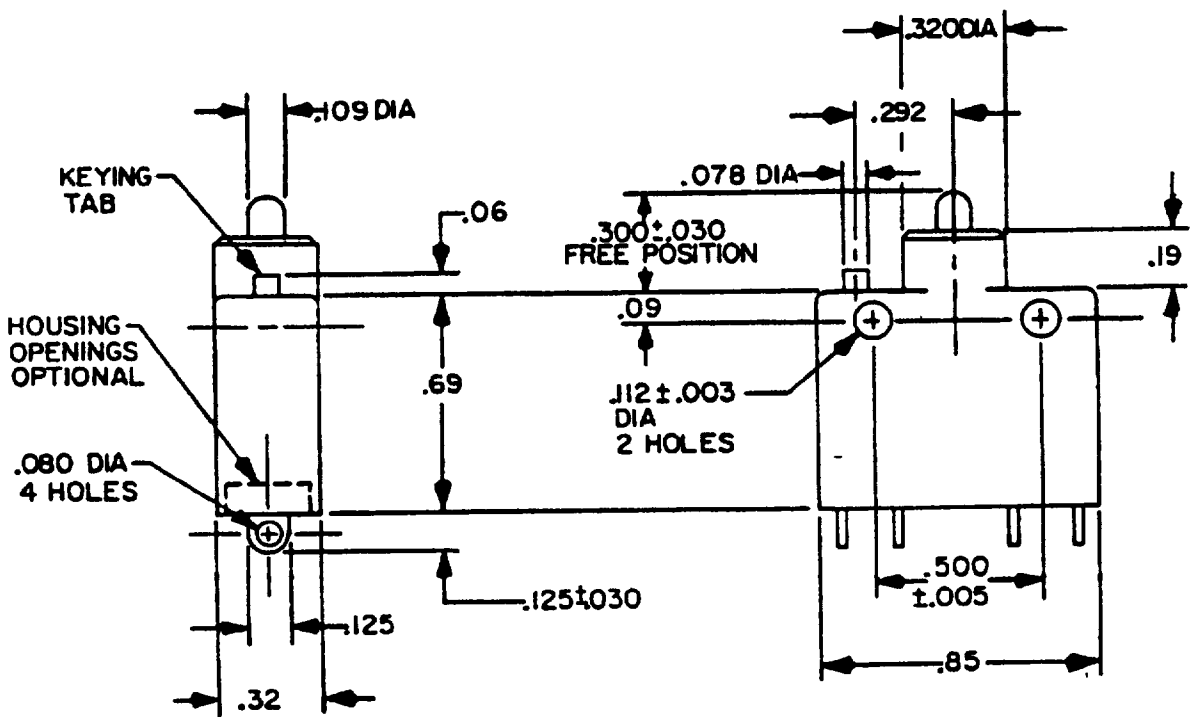
MIL-S-8805/84C
20 April 1989
SUPERSEDING
MIL-S-8805/84B
2 November 1978

MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, WATERTIGHT, 10 AMPERES, DOUBLE BREAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.



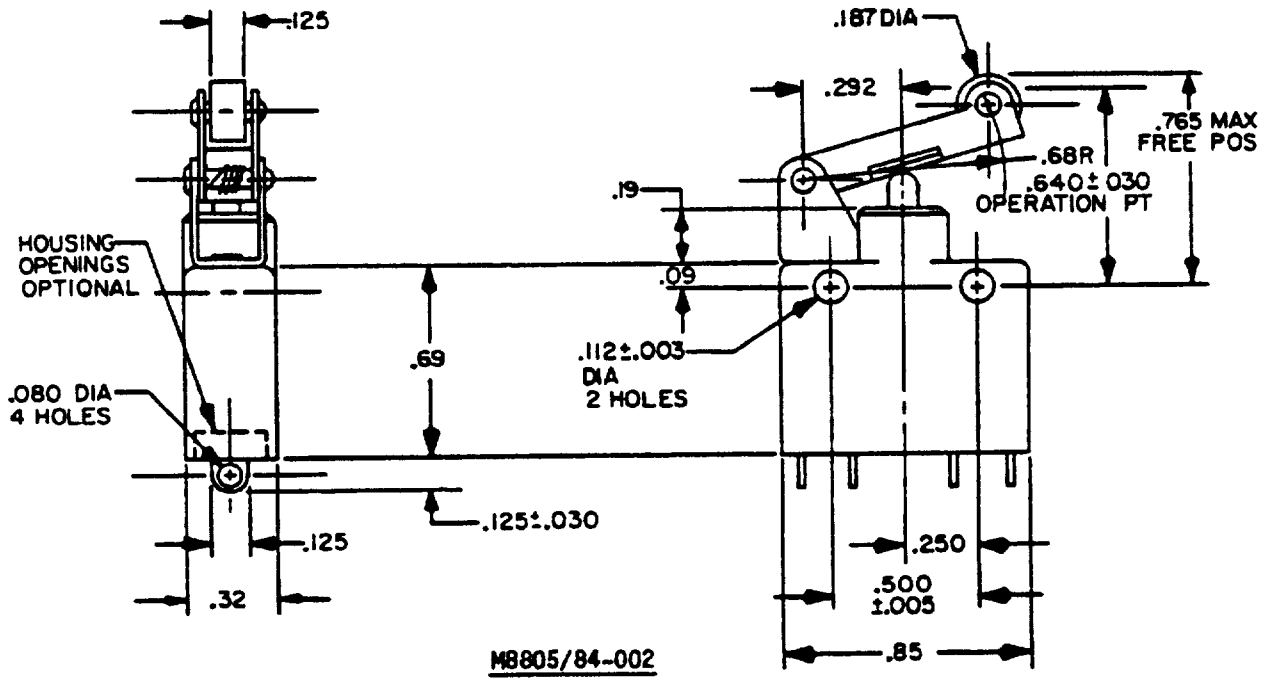
M8805/84-001

FIGURE 1. Dimensions and configurations.

© denotes changes

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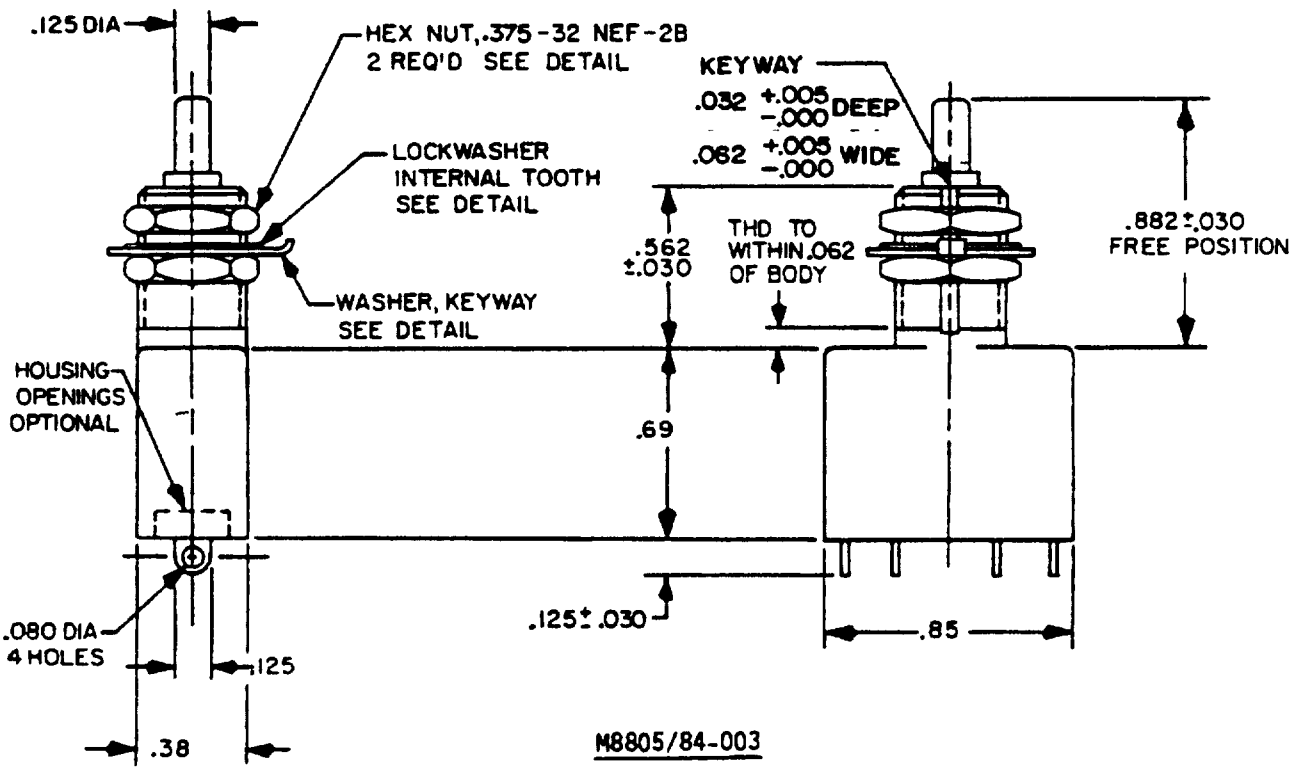
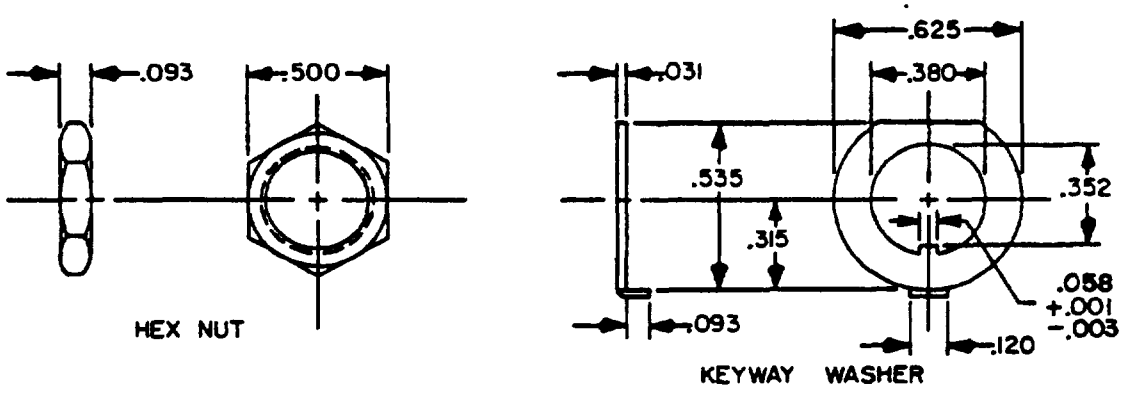
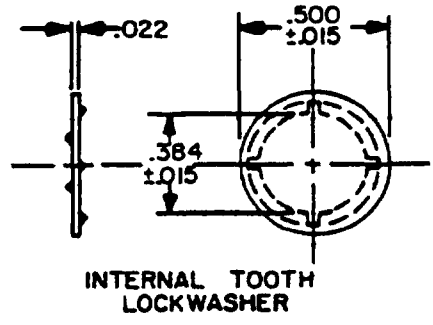


FIGURE 1. Dimensions and configurations - Continued.

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© Can be replaced by MS25082-20 or MS25082-C20



© Can be replaced by MS35333-134

M8805/84-003

FIGURE 1. Dimensions and configurations - Continued.

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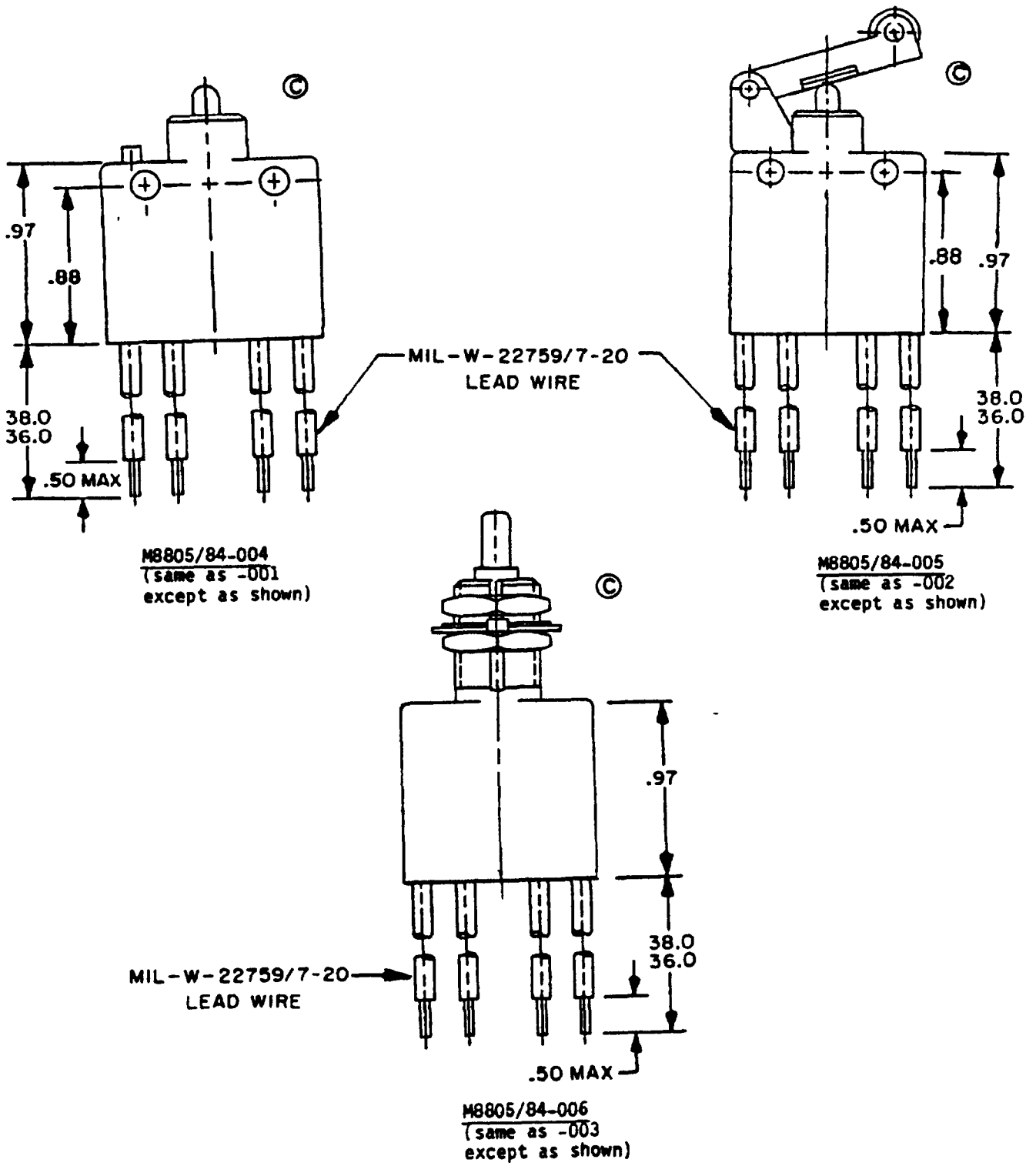
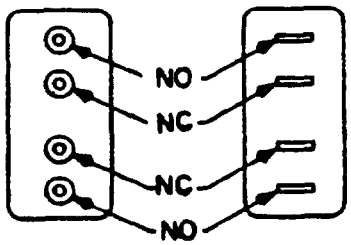
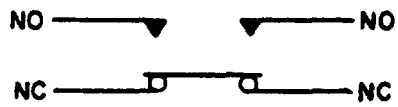


FIGURE 1. Dimensions and configurations - Continued.

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© TERMINAL IDENTIFICATION



CIRCUIT SCHEMATIC

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.06	1.5	.125	3.18	.352	8.94	.68	17.5
.003	0.08	.062	1.57	.187	4.75	.375	9.53	.69	17.3
.005	0.13	.068	1.73	.19	4.8	.380	9.65	.765	19.43
.010	0.25	.078	1.98	.250	6.35	.384	9.75	.85	21.6
.015	0.38	.080	2.03	.292	7.42	.500	12.70	.88	22.4
.02	0.5	.09	2.3	.300	7.62	.535	13.59	.882	22.40
.022	0.56	.093	2.36	.315	8.00	.562	14.27	.97	24.6
.025	0.64	.109	2.77	.32	8.1	.625	15.88	36.00	914.4
.030	0.76	.112	2.84	.320	8.13	.640	16.26	38.00	965.2
.031	0.79	.120	3.05						

NOTES:
1. Dimensions are in inches. .
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (0.5 mm) on two place decimals and ±.010 (0.25 mm) on three place decimals.
4. Terminal designation:
   M8805/84-001, M8805/84-002, M8805/84-003: NO and NC stamped on bottom or either side of switch.
   M8805/84-004, M8805/84-005, M8805/84-006: NO-20 and NC-20 stamped on leads. The marking shall be legible, permanent, shall not impair the wire characteristics, shall appear every 15 inches (381. mm) maximum, and shall appear within three inches (76.2 mm) of the end.
5. Switch design optional within envelope dimensions shown.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Enclosure design: 3 (watertight).

Temperature characteristic: 1 (-55°C to +85°C).

(C) Shock type: M (100 g's, test condition I, Method 213 of MIL-STD-202).

Vibration grade: 1 (10 to 500 Hz swept sinusoidal, 10 g's peak).

Material:

Plunger - Corrosion-resistant steel.
Lever - Corrosion-resistant steel.
Roller - Phosphor bronze.
Lockwasher and hexnut - Corrosion-resistant steel or steel, cadmium plated per QQ-P-416, or zinc plated per QQ-Z-325, type II, class 2.
Keyway washer - Corrosion-resistant steel.
Plastic - Type SDGF per MIL-M-14.

Finish: Switch housing shall be processed to resist corrosion.

Weight:

M8805/84-001 - .50 ounce maximum.
M8805/84-002 - .60 ounce maximum.
M8805/84-003 - 1.10 ounce maximum.
M8805/84-004 - 2.30 ounce maximum.
M8805/84-005 - 2.40 ounce maximum.
M8805/84-006 - 2.90 ounce maximum.

Operating characteristics: See table I.

Contact resistance (maximum):	M8805/84-001, -002, -003	M8805/84-004, -005, -006
Initial -	50 milliohms.	150 milliohms.
After mechanical endurance -	100 milliohms.	200 milliohms.

(C) Terminal strength:

Force (Static) -	5 lbs.	5 lbs.
Duration of force -	5-10 seconds.	5-10 seconds.

Mechanical endurance: 50,000 cycles.

Electrical endurance: 50,000 cycles.

Electrical ratings: See table II.

(C) Intermediate current: 50,000 cycles.

Part or Identifying Number (PIN): See table I.

Qualification:

Group submission - See table III.

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TABLE I. PIN and characteristics. 1/

PIN M8805/84	Type of actuator	Actuating force (maximum)	Releasing force (minimum)	Pretravel (maximum)	Movement differential	Overtravel (minimum)	Strength of actuator
				(inch)	(inch)	(inch)	(lbs)
-001 -004	Pin plunger	2.5 lbs	.5 lb	.050 (1.27 mm)	.020 ±.010 (.51 ±.25 mm)	.012 (.30 mm)	25
-002 -005	Roller-lever	20 oz	4 oz	.125 (3.18 mm)	.060 max (1.52 mm)	.025 (.64 mm)	10
-003 -006	Overtravel plunger	2.5 lbs	.5 lb	.070 (1.78 mm)	.020 ±.010 (.51 ±.25 mm)	.187 (4.75 mm)	25

1/ ±20 percent variation in values acceptable after test.

TABLE II. Electrical ratings.

Load	Sea level - 28 V dc		50,000 feet - 28 V dc	
	Single circuit NO or NC	2 circuit 1/	Single circuit NO or NC	2 circuit 1/
	(amperes)	(amperes)	(amperes)	(amperes)
Resistive	10	10	10	10
Inductive	7.5	5	3.5	2
Motor	4	2	---	---
Intermediate current	.04	.04	.04	.04

1/ Both NO and NC circuits under load during electrical endurance test.

TABLE III. Qualification inspection.

Examination or test	Basic switch	Other switch samples	Extent of approval
Qualification inspection table of MIL-S-8805	M8805/84-004 (all sample units)	M8805/84-001, -002, -003, -005, and -006 (2 sample units). Visual and mechanical examination and operating characteristics. M8805/84-002 and -003 strength of actuating means. M8805/84-005 vibration and shock. M8805/84-001 strength of terminals.	All

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© CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - SH
Air Force - 99
DLA - ES

User activities:
Army - AM, AT
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-1404)

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